

WILLY FITRA HENDRIA

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[willyfitrahendria](#) | [willyfh](#) | [Google Scholar](#) | [willyfh.github.io](#)

Skills

- Machine Learning | Computer Vision | Vision-Language | Full-Stack | English, Indonesian
- Python | Java | C++ | C# | JavaScript | HTML | CSS | SQL | NoSQL | Git | Docker | GCP | Airflow | FastAPI
- PyTorch | TensorFlow | Scikit-learn | OpenCV | spaCy | Pandas | TorchServe | TensorFlow Serving | MLflow

Experience

AI Research Engineer

Lunit

South Korea 07/2024 - Present

- Leading development of an internal **ML model training framework** to streamline research across teams.
- Designed and implemented a parallelized, rate-limited **LLM API pipeline**, reducing processing time from **5.6 days** to **16 minutes** for **400k+** medical text entries.
- Developed an end-to-end **downstream evaluation framework** to benchmark **foundation models** across diverse tasks and dataset.
- Integrated **timm** into the internal **training framework** and conducted **benchmarking**, enabling access to **1000+ models** and identifying a higher-performing model that improved the **mF1 score by 1.5%** over the baseline.
- Maintained internal **training and evaluation frameworks**; ranked among **top contributors** by PR count.
- **Led intern hiring and mentorship** for integrating and benchmarking **semantic segmentation** frameworks.

AI Research Engineer

Dagyeom Co., Ltd.

South Korea 10/2023 - 06/2024

- Developed a **desktop** and **web** application for the real-time anomaly detection system of video and audio data.
- Researched and deployed a state-of-the-art **anomaly detection** model (from CVPR) into an existing system.
- Implemented **multithreading** with **ONNX** in C# and Python, resulting in a **20x improvement** in inference time.
- Developed a novel approach for **unsupervised anomaly detection**, which led to a first-author publication.

AI Research Engineer

Tricubics

South Korea 04/2023 - 09/2023

- Improved multi-model ML system **FPS by 80%** via **multi-GPU** inference with **multithreading**.
- Built and deployed an **object detection** model for real-time video stream data, achieving a **35%** gain in mAP.
- Built and deployed a state-of-the-art **pose estimation** model (published in 2023), resulting in **26%** AP gain.
- Built and deployed an **image classification** model, resulting in **33%** improvement in the mF1 score.
- Improved a knowledge-based **action recognition** algorithm by integrating **object tracking** and **sensor fusion** techniques, resulting in a considerable **24%** enhancement in accuracy.

Graduate AI Researcher

VLI Lab (Sejong University)

South Korea 03/2021 - 03/2023

- Produced the highest research output in the lab by writing two first-authored papers, filing two patents, and leading teams in completing **deep learning**-based projects, such as **object detection** and **video captioning**.
- Proposed and developed a state-of-the-art **graph neural network**-based **video captioning** using **PyTorch**.
- Reproduced several **ML** papers, such as **federated learning**, **object detection**, and **video captioning/retrieval**.

Machine Learning Engineer

Detik Network

Indonesia **11/2019 - 10/2020**

- Built **Scikit-learn**-based **classification**, **regression**, and **clustering** models for AI use cases in online media articles, such as **tagging** and **categorization** of articles, **recommendation engine**, and **customer segmentation**.
- Improved preprocessing time of 20 million data from 50 minutes to 3 minutes by converting the **Spark**-based code to **Google BigQuery**.
- Built monitoring and inference pipeline of **Google AutoML** model for batch prediction by using **Apache Airflow**.

Software Engineer **Works Applications Co., Ltd.** *Singapore, Japan* **01/2016 - 09/2018**

- **Led a 3-member team** to develop a UI component library, including implementation, code reviews, and testing.
- Improved build time of a large project from more than 1 hour to less than 20 minutes.
- Designed and developed the **front-end** and **back-end** of an e-commerce system using **Java Spring** framework.

Education

Master of Science **Sejong University** *South Korea* **03/2021 - 02/2023**

- Major in Intelligent Mechatronics Engineering and Convergence Engineering for Intelligent Drone (Dual Degree).
Thesis: Video Captioning Based on Graph Neural Networks Using Action Knowledge

Nanodegree **Udacity** *Online* **11/2019 - 02/2020**

- Program in Machine Learning Engineer. **Capstone Project:** Cat Breed Image Classification Using CNN [\[link\]](#)

Master of Science (Incomplete) **The University of Bonn** *Germany* **04/2019 - 09/2019**

- Major in Computer Science. **Courses:** Advanced Deep Learning for Graphics, Data Mining and Knowledge Discovery, Robot Learning, Humanoid Robotics, Artificial Life

Bachelor of Science **Bandung Institute of Technology** *Indonesia* **08/2011 - 09/2015**

- Major in Computer Science. **Relevant Courses:** Artificial Intelligence, Machine Learning, Software Engineering.

Projects

- **Lightning Hydra Boilerplate:** Deep learning experiment template using PyTorch Lightning + Hydra [\[link\]](#) (2025)
- **VisualTorch:** A tool for visualizing PyTorch-based neural network architectures [\[link\]](#) (2024)
- **MSVD-Indonesian:** A benchmark for multimodal video-text tasks in Indonesian [\[link\]](#) (2023)
- **Graph Transformer:** An unofficial implementation of Graph Transformer in PyTorch [\[link\]](#) (2023)
- **CLIP4Caption:** An unofficial implementation of CLIP4Caption in PyTorch [\[link\]](#) (2022)

Selected Publications

- Multi-model anomaly detection for industrial inspection with dynamic loss weighting and soft-hard features loss, Neural Computing and Applications (Q1), Springer [\[link\]](#) (2025)
- A multilingual multimodal data hub and benchmark suite for Southeast Asian languages, EMNLP [\[link\]](#) (2024)
- Action knowledge for video captioning with graph neural networks, JKSU-CIS (Q1), Elsevier [\[link\]](#) (2023)
- Non-contact supervision of COVID-19 breathing behaviour with FMCW radar and stacked ensemble learning model in real-time, Transactions on Biomedical Circuits and Systems (Q1), IEEE [\[link\]](#) (2022)
- Combining transformer and CNN for object detection in UAV imagery, ICT Express (Q1), Elsevier [\[link\]](#) (2021)

Patents

- Knowledge distillation for graph-based video captioning, KR, **Granted**, No. 102799596 [\[link\]](#) (2025)
- Transfer of tactile data in teleoperation system, KR, **Granted**, No. 102611269 [\[link\]](#) (2023)

Others

- Contributed to several open-source projects, including [Anomalib](#), [MMPose](#), and [SEACrowd](#) (2023 - 2024)
- Served as the Sejong University **representative** for the Indonesian Students Association in Korea (2021 - 2023)
- Spoke as an **invited speaker** on “Object Detection for Drone Imagery” in the digiXed Webinar (03/2023)
- Awarded the **AWS Scholarship**, top 300 in the Amazon Web Services DeepRacer Challenge (11/2019)
- Achieved a **Gold Medal** in an intra-university chess team competition at Olympiad VIII KM-ITB 2015 (02/2015)